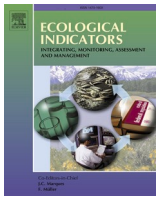
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Urban heat islands and their effects on thermal comfort in the US: New York and New Jersey

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ABSTRACT

In recent decades, urbanization and the dramatic increase in urban populations have exacerbated the urban heat island effect. At present, much attention has been paid to the causes and patterns of the urban heat island effect. However, there are few quantitative studies on the impact of the urban heat island effect. By extracting post-world climate data, DEM data, and land use data for 20 years between 2001 and 2020, this paper first studies the variation pattern and spatial distribution characteristics of urban heat islands in New York State, summarizes the seasonal distribution characteristics of temperature, and then uses Giles formula to calculate Tom's discomfort index and evaluate human thermal comfort, and evaluates the effect of heat island effect on human sensory thermal comfort. The results show that on the time scale, the surface temperature in the study area generally showed a slow upward trend over the past 20 years. For example, in July, the maximum and minimum temperatures increased by 3.2 °C and 4.1 °C, respectively. At the spatial scale, most of the heat island areas in the study area were distributed in the New York City agglomeration, especially from May to October, when the heat island effect was particularly obvious. The temperature map showed obvious high temperatures. Compared with 2001, the human discomfort index (DI) increased between June and August 2020. The land use map shows that as the city expands, people's DI index also increases, and the proportion of people who feel uncomfortable with heat increases to 50%. Except for the New York City cluster, other areas were mostly athermic-free. This result shows that excessive urban development concentration seriously affects residents' quality of life. We should pay attention to the superimposed impact of climate change and urban heat islands on the human discomfort index, and adjust the local high temperature and thermal field area through reasonable planning, strengthening greening, and using building technology to make cities more livable.

1. Introduction

Accompanying the rapid development of global urbanization in recent decades, increasing deterioration of the natural environment has been widely observed. Among all the environmental problems brought by urbanization and industrialization, the phenomenon of urban heat islands has attracted scholars' interest for a long time (Wang et al., 2019). The phenomenon that the temperature in the center of the city is higher than the temperature in the suburbs was first proposed by Howard in 1818(Rahman et al., 2022). In 1968, Bornstein proved that the heat island effect is real in New York City, USA (Bornstein, 1968). Since then, many experts and scholars have begun conducting empirical studies on urban heat islands' characteristics. Especially after the maturity of remote sensing (RS) and geographic information system (GIS) technology research, the combination of Landsat remote sensing data with ENVI and ArcGIS helped to extract the image and spatial mapping, make the data visualization and solve the problem of low accuracy of previous research and can be deeply combined with urban heat island research and actual investigation (Almeida et al., 2022; Chen et al., 2006; Kolokotroni et al., 2006). For example, Dwivedi and Mohan used Landsat remote sensing data combined with ENVI and ArcGIS software to study the overall thermal field phenomenon in Mumbai, India (Dwivedi and Mohan, 2018). Much attention has been paid to the causes and patterns of the urban heat island effect. The results show that

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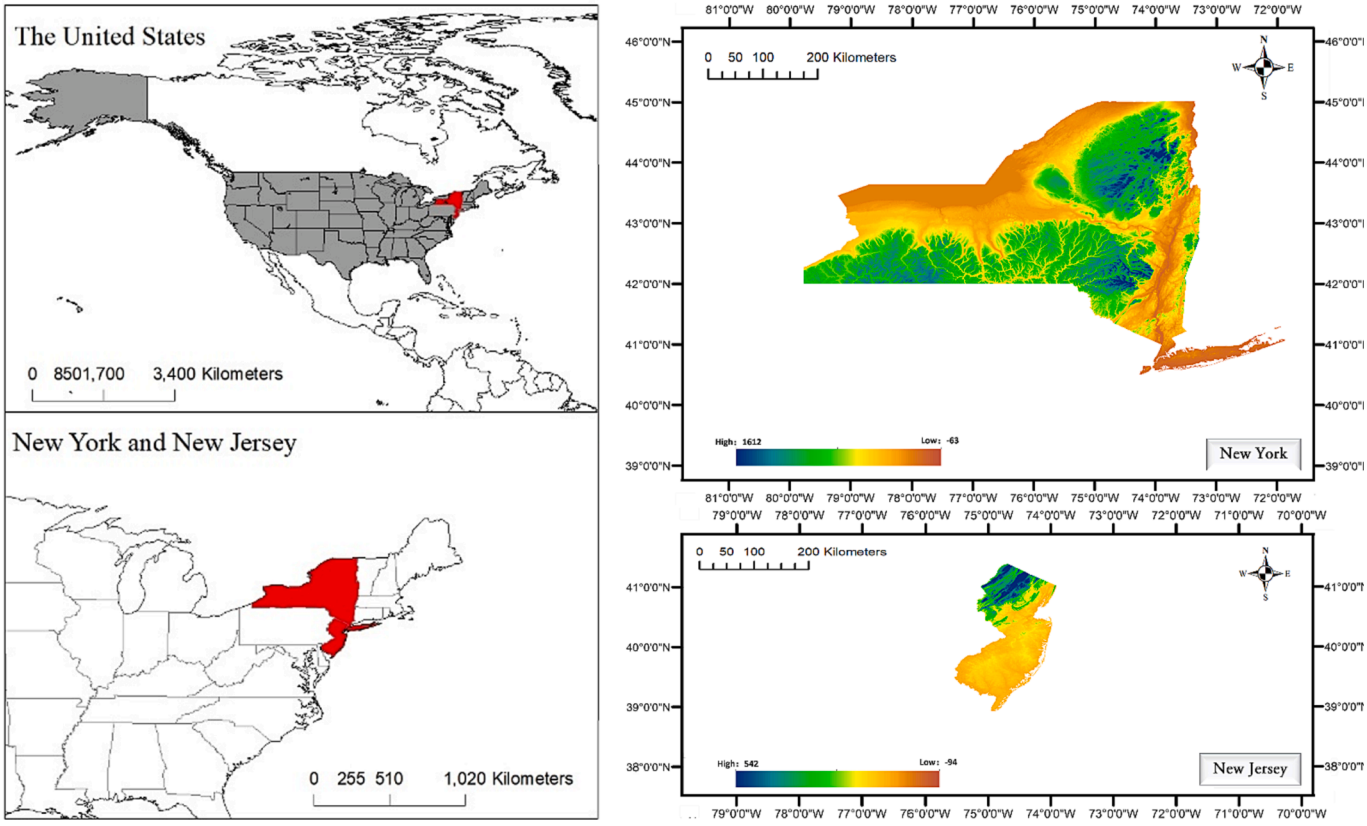


Fig. 1. Distribution of study area.

for the city as a whole, the uneven expansion and construction of the city will lead to an uneven temperature growth rate in the corresponding area(Jauregui et al., 1992). For the spatial distribution characteristics of cities, different land use types such as opaque land surface, vegetation and water distribution, as well as the absorption of long-wave radiation in *peri*-urban areas, and the significant increase in the temperature difference of the underlying surface, all of which will lead to the formation of the heat island effect(Cheela et al., 2021; Feyisa et al., 2014; Liu et al., 2014; Yao et al., 2022).

Researchers started to realize and prove that the heat island effect will affect the ecological environment, cause air pollution (Sarrat et al., 2006), and to a certain extent, affect the comfort of people's bodies and the comfort of residents' lives. However, most existing studies focus on the phenomenon of surface temperature increase caused by the heat island effect and rarely quantify the impact of the urban heat island effect on human subjective temperature perception. The human body can directly feel changes in thermal temperature and reflect them as comfortable or uncomfortable sensations when the temperature changes. A survey of heat mortality rates in New York City found that the highest heat index performed similarly to other more complex measures when estimating the risk of death during hot weather. This also shows that temperature can be used as a measure of health risks. (Metzger Kristina et al., 2010).

In order to study the influence of the heat island effect on human comfort, this study explores the change patterns and spatial distribution characteristics of urban heat islands in New York State. In addition, it evaluates its impact on the human body's thermal comfort. New York State is one of the most economically developed states in the United States. Therefore, it is facing obvious urban heat island problems. We apply Thom's Discomfort Index, alternated by Giles et al. in 1990, to evaluate human comfort (Giles et al., 1990) and make suggestions for future urban management.

Discomfort Index was proposed in the 1920 s and developed for a long time. In 1923, Houghton proposed an Effective Temperature Index that profoundly impacted thermal comfort research (de Freitas and Grigorieva, 2017). In 1957, the U.S. Navy proposed the famous Wet Bulb Globe Temperature Index to prevent thermal damage accidents in military training. European and American countries have also used this index to evaluate high-temperature workshops' hot environment meteorological conditions for many years. However, it is unsuitable for assessing comfort or cold climate (Yaglou and Minaed, 1957). In 1959, Thom of the National Weather Service proposed the Discomfort Index, also later referred to as a Temperature-Humidity Index, which reflects the influence of the hot and humid climate environment on human discomfort in summer through the combination of dry bulb temperature and wet bulb temperature and is applied to the U.S. summer comfort and working hours forecast (Thom, 1959). This index was empirically obtained by subjecting several samples of people to varying temperatures and humidities. Then they filled out a questionnaire about their comfort based on their feelings about their environment. In 1979, based on several previous biometeorological studies, R. G. Steadman published a new Temperature–Humidity Index (Steadman, 1979). Unlike the previous Discomfort Index, this index indicated the apparent temperature that a person would feel relative to a "normal" humidity corresponding to a dew point of 14 °C. Steadman's model was based on five submodels, some of which had been empirically obtained. He also identified numerous parameters such as a person's build, activity, clothing, solar radiation, and effective wind speed, all of which influenced a person's perception of apparent temperature. In 1984 Steadman refined his model and included the effect of wind chill to produce a "universal scale of apparent temperature" (Steadman, 1984). In 1990, Giles et al. refined and proposed an alternative version of Thom's DI, defined in terms of temperature (degrees Celsius) and relative humidity (percentage)(Giles et al., 1990).

Given the high correlation between temperature and comfort, we believe that the effects of urban heat islands and other factors on human

comfort can be observed by studying temporal and spatial changes in temperature. The structure of this paper is as follows: Section 2 describes the study area, data sources, and methods used. Section 3 analyzes the influence of land surface temperature changes and land use, geomorphology, etc., on thermal comfort at the spatiotemporal scale, Section 4 discusses possible methods to improve the discomfort index and the shortcomings of this paper, and Section 5 restates the findings of this paper and discusses the future.

2. Materials and methods

2.1. Materials

2.1.1. The study area

This study area is shown in Fig. 1, New York and New Jersey in the US. New York State covers an area of 128,401 square kilometers and is located in the northeastern United States. New York is a state in the Mid-Atlantic and Northeastern regions of the United States. New York State is the nerve center and economic heart of the United States. Its finance, commerce, industry, art, clothing, etc., occupy leading positions in the U.S. states. It has the largest city in the United States, New York City, and New York Harbor. Two-thirds of the state's population lives in the New York metropolitan area, one of the most populous in the world. With a population of 8,804,190 in 2020, New York City is the most populous city in the United States. In addition, New York City is relatively close to New Jersey, and a large part of New Jersey belongs to the New York metropolitan area. Therefore, New Jersey is included in the research area to better study the relationship between New York's urban heat island phenomenon and physical comfort. The selection of New York and New Jersey is of great significance and research value for studying the heat island effect in the United States.

Topographically, the St. Lawrence Lowlands are located in the northern part of the state, stretching from Alexandra Bay to the west and bordering the St. Lawrence River with Canada. The Ediron Deckard High region is located northeast of New York State, with the Ediron Deckard Mountains. The Great Lakes Low, northwest of New York State, is bordered by Lake Erie and Lake Ontario. The Hudson-Mohaq Low is located in the central and southeastern parts of New York State, including the valley of the Hagopan-right Mohawk River. More than half New York's southeastern border belongs to the High New England region. The area stretches from Lake Champlain to Manhattan Island. The Atlantic Coastal Plain includes Stanton Island, Long Island, and the coastal area.

Climatically, the study area has a humid continental climate in general. Weather in New York is heavily influenced by two continental air masses: a warm, humid one from the southwest and a cold, dry one from the northwest. Downstate New York, comprising New York City, Long Island, and lower portions of the Hudson Valley, has rather hot summers with some periods of high humidity and cold, damp winters, which are relatively mild compared to temperatures in Upstate New York due to the downstate region's lower elevation, proximity to the Atlantic Ocean, and relatively lower latitude. According to the study, New York's climate change is attributed to increased carbon dioxide emissions as the city develops. It is of concern due to its impact on the state's people, ecosystem, and economy. Many parts of the state are already experiencing weather changes and sea-level rise, threatening local communities.

2.1.2. The data

Based on the research objectives, climate data, land-use data, and the digital elevation model were selected for experimentation in this paper.

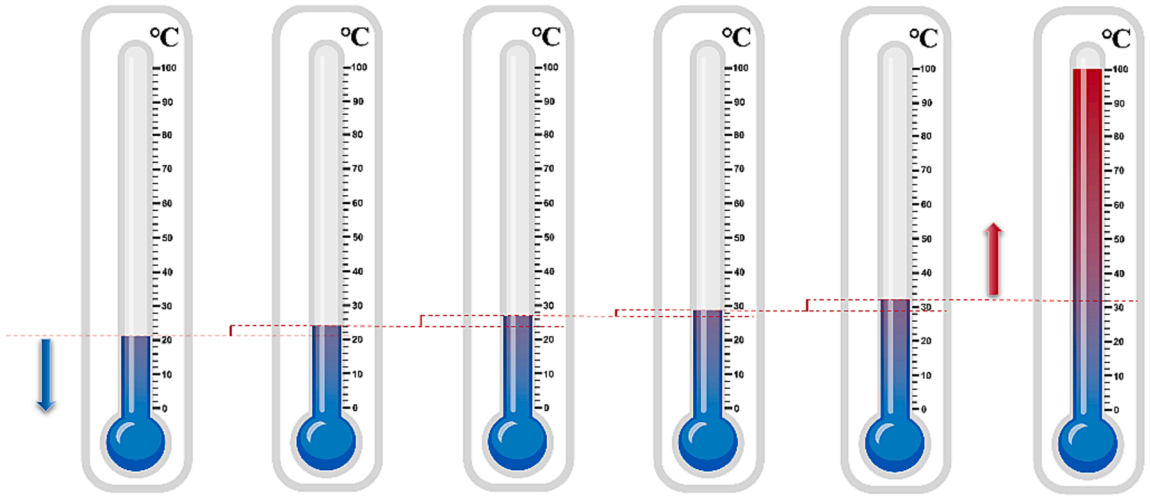
The first category is climate data, including surface temperature, relative humidity, wind direction, and wind speed. The main dataset is surface temperature data, an important input factor for various plant physiological, hydrological, meteorological, environmental, and other near-surface meteorological parameters in various models (Prihodko and Goward, 1997). Global surface temperature information is usually derived from the National Aeronautics and Space Administration (NASA)'s Goddard Institute for Space Research, the National Oceanic and Atmospheric Administration's National Climatic Data Centre (NCDC), the MET Office's Hadley Centre, and the University of East Anglia Climatic Research Unit (HadCRUT) (Brohan et al., 2006). The most common method for obtaining station climate data is spatial interpolation. However, due to the unevenness of the weather stations, they always make large errors (Jin and Dickinson, 2010; Mildrexler et al., 2011). Therefore, Fick and Stephen created a new spatially interpolated monthly climate dataset that includes monthly temperatures (minimum, maximum, and average), precipitation, solar radiation, vapor pressure, and wind speed. The weather station data was interpolated using thin-plate splines with covariates including elevation, distance to coast, and three satellite-derived covariates: maximum and minimum land surface temperatures and cloud cover, obtained by the MODIS satellite platform (Fick and Hijmans, 2017). The spatial data (https://www.worldclim.org/) of month-by-month average surface temperature, relative humidity, and average wind speed in New York and New Jersey from 2001 to 2020 are used in this study.

Table 1
Land use type table.

Name	Vale	Description
Evergreen Needleleaf Forests	1	Dominated by evergreen conifer trees (canopy > 2 m). Tree cover > 60%.
Evergreen Broadleaf Forests	2	Dominated by evergreen broadleaf and palmate trees (canopy > 2 m). Tree cover > 60%. [1]
Deciduous Needleleaf Forests	3	Dominated by deciduous needleleaf (larch) trees (canopy > 2 m). Tree cover > 60%.
Deciduous Broadleaf Forests	4	Dominated by deciduous broadleaf trees (canopy > 2 m). Tree cover > 60%.
Mixed Forests	5	Dominated by neither deciduous nor evergreen (40–60% of each) tree type (canopy > 2 m). Tee cover > 60%.
Closed Shrublands	6	Dominated by woody perennials (l-2 m height) > 60% cover.
Open Shrublands	7	Dominated by woody perennials (l-2 m height), 10–60% cover
Woody Savannas	8	Tree cover 30–60% (canopy > 2 m).
Savannas	9	Tree cover 10–30% (canopy > 2 m).
Grasslands	10	Dominated by herbaceous annuals (<2m).
Permanent Wetlands	11	Permanently inundated lands with 30–60% water cover and > 10% vegetated cover.
Croplands	12	At least 60% of the area is cultivated cropland.
Urban and Built-up Lands	13	At least 30% impervious surface area, including building materials, asphalt, and vehicles.
Cropland/Natural Vegetation Mosaics	14	Mosaics of small-scale cultivation 40–60% with natural tree, shrub, or herbaceous vegetation.
Permanent Snow and Ice	15	At least 60% of the area is covered by snow and ice for at least 10 months of the year.
Barren	16	At least 60% of the area is non-vegetated barren (sand, rock, soil) areas with less than 10% vegetation.
Water Bodies	17	At least 60% of the area is covered by permanent water bodies.
Unclassified	255	Has not received a map label because of missing inputs.

The second category is land-use data. Land cover plays an important role in the climate and biogeochemistry of the Earth system. This paper selects the MCD12Q1 satellite data of the study area from 2001 to 2020 as the basic urban land use data. Five different land cover classification schemes were adopted in the MCD12Q1 satellite. The type1, the global vegetation classification scheme of the International Geosphere-Biosphere Programme (IGBP), was adopted in this study. The specific types are shown in Table 1.

The third category is the digital elevation model, a data structure commonly used to store terrain information, usually interpolated to establish the value of the entire terrain point. DEM can be obtained from contours, topographic maps, field surveys, photogrammetry techniques,



	1	2	3	4	5	6
DI(°C)	<21	21-24	24-27	27-29	29-32	>32
People's feel	No discomfort	Under 50% population feels discomfort	Over 50% population feels discomfort	Most of population suffers discomfort	Everyone feels severe stress	State of medical emergency

Fig. 2. Values of Thom's Discomfort Index (DI) calculated for Celsius temperature scales.

radar interferometry, and laser measurement high schools (Saleem et al., 2019). The data in this article is primarily downloaded from USGS (https://earthexplorer.usgs.gov/).

2.2. Methods

2.2.1. Data Pre-processing

Data pre-processing includes data extraction, image processing, and improving the accuracy of meteorological station data. In addition, it should provide a concise and precise description of the experimental results, their interpretation, and the experimental conclusions that can be drawn.

Data extraction.

In this paper, we mainly process the data in two steps. The first step is data extraction, and the second is image processing. The data downloaded from ' WorldClim 2′ are spatial data for data extraction. Since their attribute data are needed in the subsequent data processing statistics, the attributes of spatial data need to be extracted in advance. The data of the past 20 years are selected and processed for the calculation to create conditions for later study of the factors influencing the urban thermal field.

Image processing.

In image processing, the posterior, land use, and DE data were stitched, corrected, and cropped by ArcGIS 10.3 software in this study. In addition, the study area's New York and New Jersey boundaries were extracted from the vector data of the World Administrative Region in ArcGIS10.3 software.

Improving the accuracy of meteorological station data.

Jin et al. proposed Estimates for regions where weather station density (or quality) is insufficient to resolve local gradients that may be improved using remotely sensed meteorological data from satellites (Jin and Dickinson, 2010). In previous studies, interpolating meteorological data from meteorological stations suffered from large errors due to the uneven density of meteorological stations. Jin et al. suggested that a meteorological station's lack of density (or quality) could be solved using remotely sensed satellite meteorological data (Jin and Dickinson, 2010). With advances in satellite technology, many meteorological variables, such as land surface temperature (LST; the Earth's "skin temperature") and cloud cover, are now available from many satellites. And created global monthly precipitation, minimum, mean, and maximum temperatures, solar radiation, wind speed, and relative humidity. Our surfaces have a 30 arc-second spatial resolution; this is equivalent to about 0.86 km2 at the equator, and the data are referred to as the ' WorldClim 2′ database and are available for download from the '1-km' spatial resolution. Database and are available for download from https://worlclim.org/. This study is mainly based on their monthly mean temperature and relative humidity data (Fick and Hijmans, 2017).

2.2.2. Heat island effect calculation and visualization

Kriging interpolation is a spatial interpolation method developed based on the variogram model, which is widely used in geology, soil, agriculture, meteorology, and other fields and is mainly used for studying spatial differentiation and mapping (Matheron, 1963). The calculation of the semivariogram and the fitting of the theoretical model in this paper are completed by the spatial interpolation module in the geostatistics software ArcGIS.

In order to gain an in-depth understanding of the changes in time and space of surface temperatures in New York and New Jersey, this paper selects the scenes of 12 months per year from 2001 to 2020. It interpolates the temperature data of New York and New Jersey for 12 months based on ArcGis10.3. In order to analyze the changes in surface temperature more intuitively and accurately, this study counts the minimum, average, and maximum values of the surface temperature, takes the average value as the surface temperature change standard, and takes the maximum and minimum values as the surface temperature difference analysis.

2.2.3. Calculation of human thermal comfort

Although various studies have given formulas for calculating thermal comfort, they are all close to the thermal index. However, there is no equation for calculating the heat index directly. Instead, heat Index values are derived from a collection of equations that comprise a model (Coccolo et al., 2016). Human thermal comfort in urban areas provides information to the city dwellers and the urban planner about the adverse

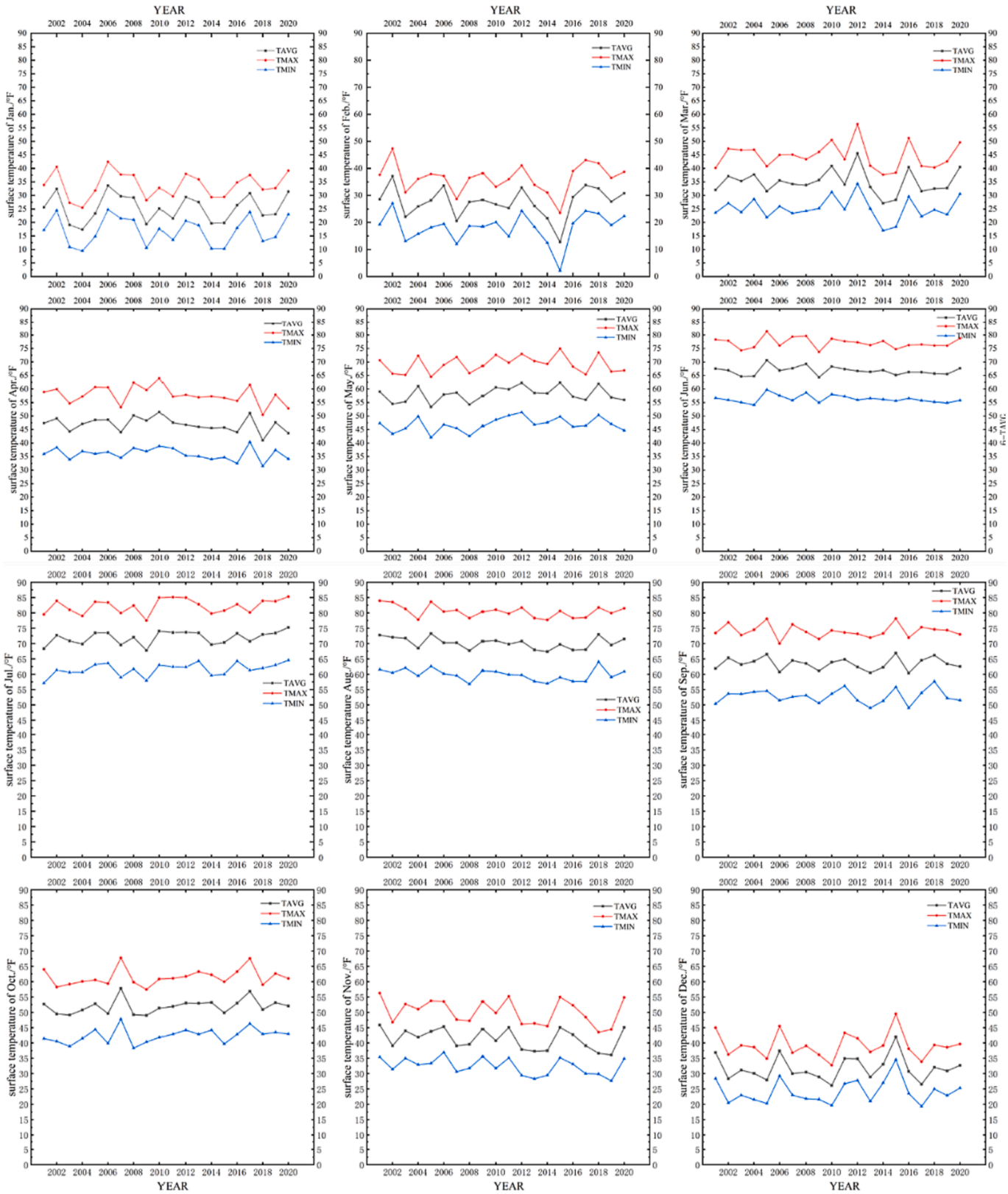


Fig. 3. Statistics of temperature minimum, average, and maximum from 2001 to 2020, from January to December.

impacts on human health due to extreme temperatures and increased Land Surface Temperature. Thom proposed Discomfort Index, a simple and widely used index in calculating thermal comfort (Thom, 1959). Based on Thom's research, the formula of Giles (Giles et al., 1990) was analyzed, and the formula is as follows:

$$DI = Ta - 0.55*(1 - 0.01*RH)*(Ta - 14.5) \quad (1)$$

Ta is ambient dry bulb temperature (℃), and RH is relative humidity (%). In this study, we used land surface temperature instead of Ta, for the two have a strong correlation.

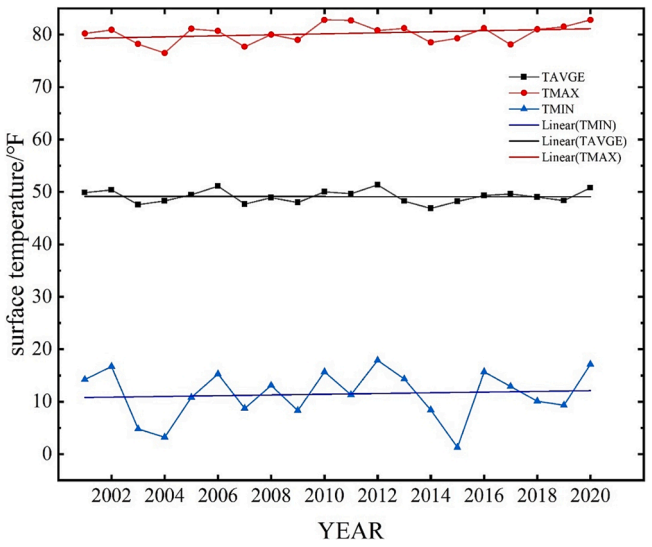


Fig. 4. Comparison of the changing surface temperature trend by the average, maximum, and minimum values from 2001 to 2020.

Giles divided the comfort level, as shown in Fig. 2.

3. Results

3.1. Spatial-temporal distribution of surface temperatures

Spatial-temporal analysis of surface temperature allows you to understand the spatiotemporal variation of temperature and study possible influencing factors. This paper analyzes the changes in surface temperature from two dimensions: time and space.

3.1.1. Temporal distribution of surface temperatures

In order to be able to deeply understand the changes in time and space of surface temperature in New York and New Jersey, this paper selects the monthly average temperature and maximum and minimum values of each month from 2001 to 20 years for interannual comparison. The results are shown in Fig. 3. It can be seen that the changing trend between the average, maximum, and minimum values of each month in the past 20 years is the same, indicating that the average temperature data is more reliable. For example, the highest surface temperature value in 20 years was 29.6℃, which appeared in July 2020, and the lowest in February 2015, which was −27℃. However, the trend of temperature changes in each month over the years was not the same. Although the change was large, some months, such as June temperature data, changed more moderately in 20 years, but some months, such as February, had great fluctuations in the past 20 years, so it showed that the changes in surface temperature in different months over the past 20 years were different.

In order to analyze the specific changes in surface temperature in the study area over the past 20 years more intuitively and accurately, this study statistically and summarizes the data of New York and New Jersey for each month from 2001 to 2020, which allows us to more intuitively observe and analyze the changes in the surface temperature of the study area. The results are shown in Fig. 4.

Fig. 4 compares the average, maximum, and minimum annual temperatures observed at the New York and New Jersey Weather Stations from 2001 to 2020. It can be found that the highest temperature and lowest temperature are all on the rise. However, there has been a major cooling in 2015–2018, and the overall trend is still rising. As a result, the

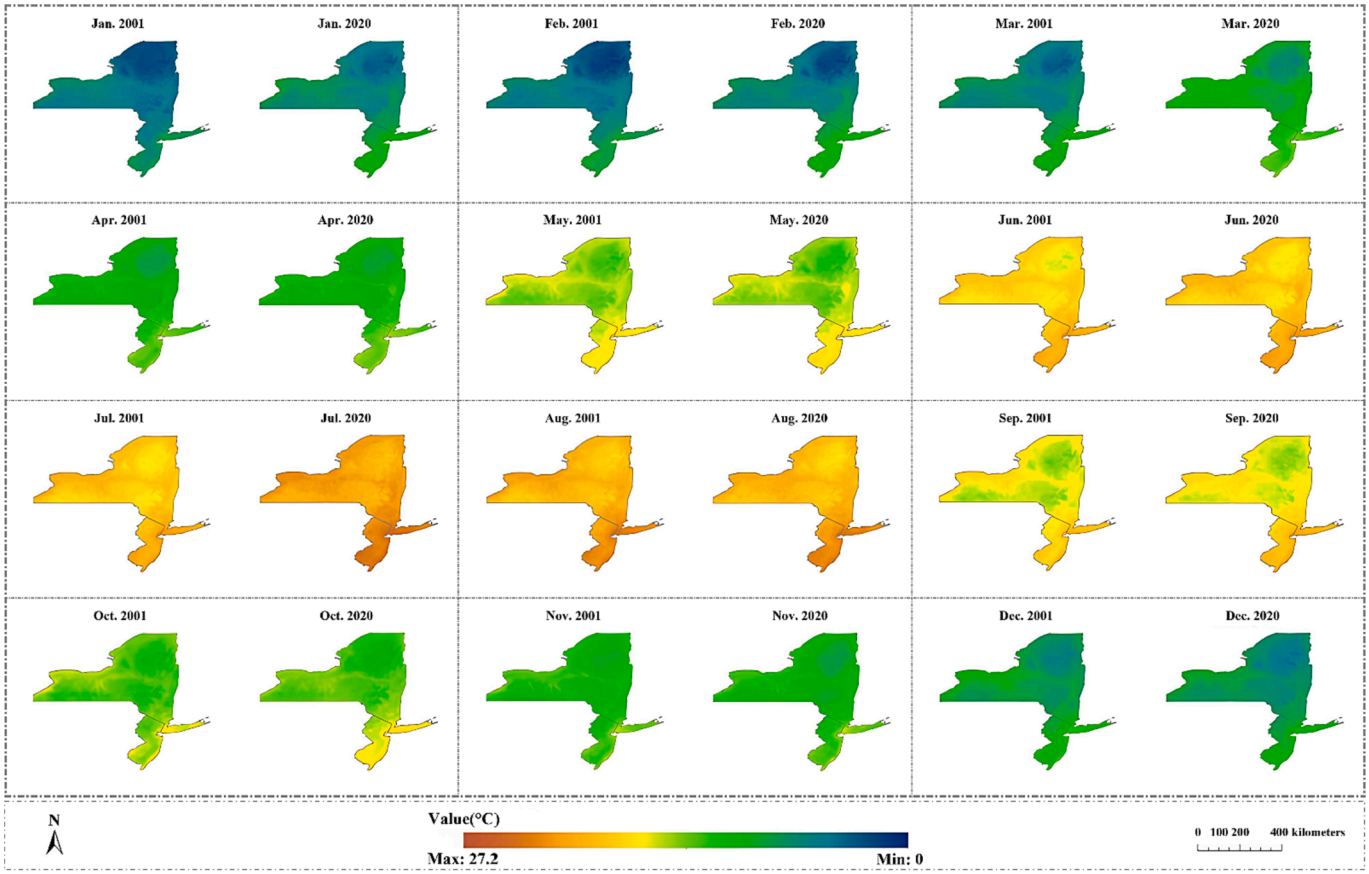


Fig. 5. Surface temperature comparison maps between 2001 and 2020, from January to December.

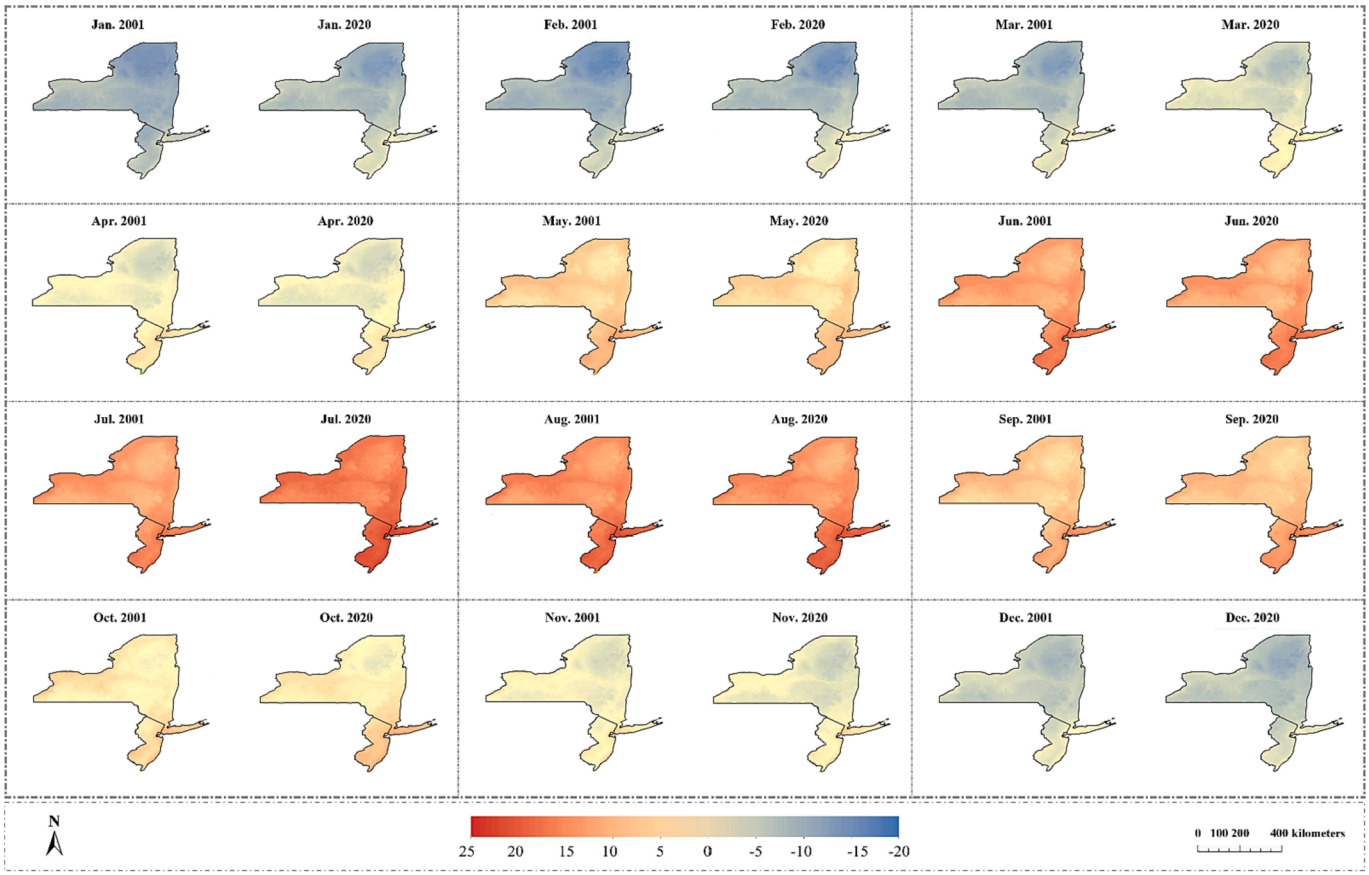


Fig. 6. Comparison maps of comfort indices for each month from January to December 2001 and 2020.

average temperature change trend is small, and the curve is relatively stable. However, the rising trend of high temperatures indicates that the heat island effect and the enhancement of climate change will increase the discomfort index of the human body.

3.1.2. *Spatial distribution of surface temperatures*

Taking New York and New Jersey as the study scope, the interpolation method calculated the surface temperature results. As a result, 24 pairs of temperature data in 2001 and 2020 were obtained, as shown in Fig. 5.

As seen in Fig. 5, the temperature in 2001 showed a general trend of gradually becoming hotter from south to north. However, in part, the temperature in most parts of the southern border and central parts of New York State is lower than that of other regions. For example, the highest surface temperature is 24.2℃ in June. The high-temperature concentration area is in the New York City Cluster at the intersection of New York State and New Jersey, of which the surface high-temperature area from June to August shows a surface-like distribution; the lowest temperature is −12.0℃. And the low temperature is speckled, mostly concentrated in the image center area.

The maximum surface temperature in 2020 is 26.8℃. The surface high-temperature area shows a surface-like distribution. The overall temperature of the New York urban agglomeration from June to August is higher than in 2001. However, the average temperature of the surface in January, February, and December 2020 was lower than in 2001. As a result, the temperature difference between the whole year of 2020 and the temperature difference is greater than in 2001, which attached 10℃. After combining local temperature monthly results from 2001 and 2020, it can be found that regardless of 12-month changes in surface temperature, the New York urban agglomeration remains a concentrated area of relatively high temperatures and is speckled, especially from June to September; the lower temperatures in the north are since the area is mostly lake and forest green.

Fig. 5 also compares the map of the surface temperature zone for each month of 2001 and 2020 and shows a relatively clear overall temperature increase trend, especially from May to 1.0. During the month, the temperature of the New York urban agglomeration in 2020 was significantly higher than in 2001. This also indicates the strengthening of the heat island effect, justifying the increase in the human discomfort index.

3.2. *Temporal and spatial distribution of somatosensory comfort levels*

Fig. 6 compares the comfort indices for different months in 2001 and 2020. In January and February 2001, most of the study area will be in a state of cold discomfort, but in the vicinity of the New York City Cluster. In the spring of April, May, and October 2001, the heat island phenomenon was concentrated in the southeast of the image, that is, the location of the urban agglomeration, but even in the strong heat island effect, the human body still feels more comfortable. From June to August 2020, the phenomenon of strong heat island and heat island was concentrated in the central area of the image, corresponding to the feeling of human heat, heat, and heat, even if the human body in the middle-temperature area is still in a slightly hot and hot feeling, people in these places in the summer, the body is a state of heat discomfort. In 2020, the overall somatosensory index improved compared with 2001, the DI index in June-August was enhanced, and the heat island effect was more obvious.

3.3. *Factors associated with urban heat fields*

In the process of accelerating the urbanization process, the types of

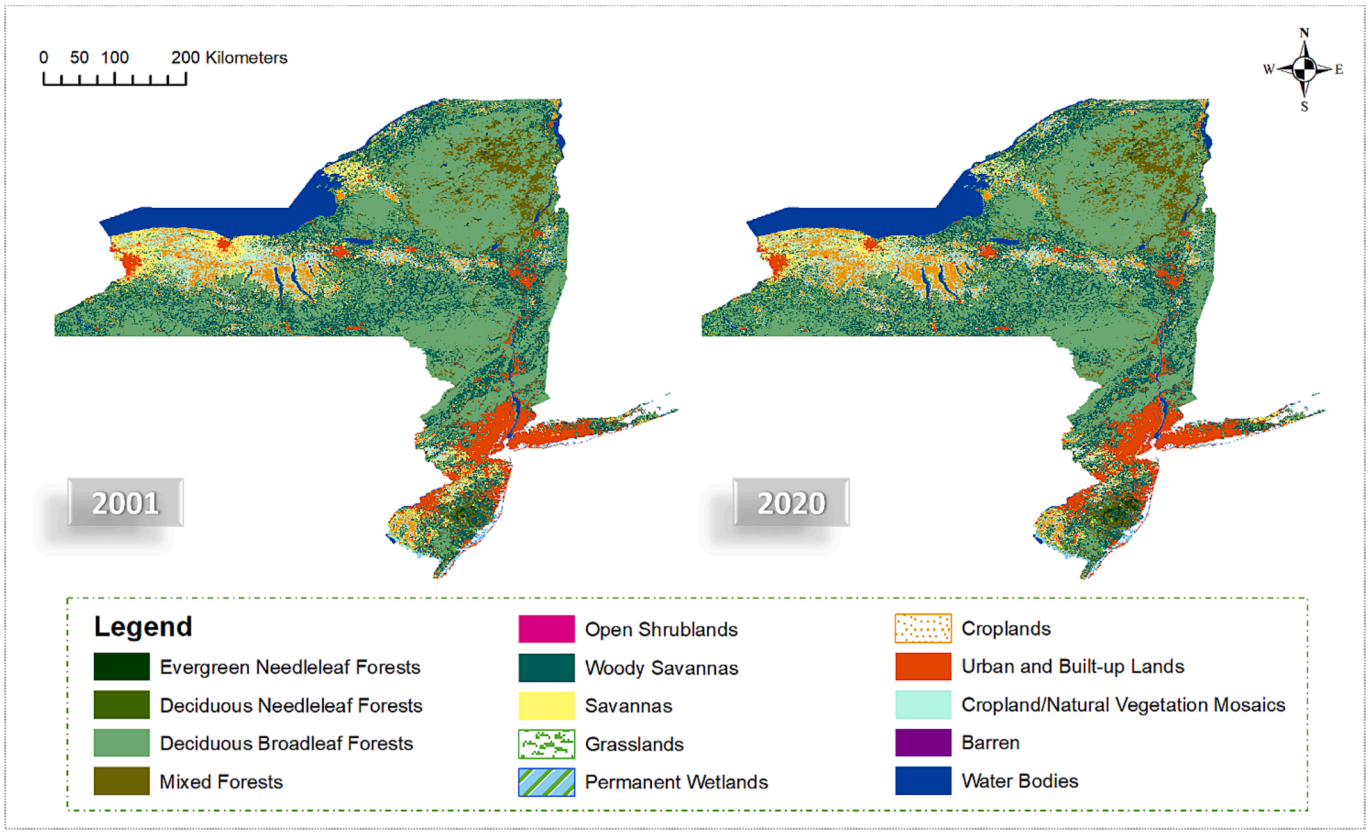


Fig. 7. Land use types in New York and New Jersey in 2001 and 2020.

Table 2
Comparison of land use type area data and change between 2001 and 2020.

value	Landuse type	Count (2001)	Count (2020)	Change (Count)	Change (%)
1	Evergreen Needleleaf Forests	1750	2894	1144	65.37
3	Deciduous Needleleaf Forests	68	192	124	182.35
4	Deciduous Broadleaf Forests	228,421	220,247	−8174	−3.58
5	Mixed Forests	30,442	31,161	719	2.36
7	Open Shrublands	1	0	−1	−100.00
8	Woody Savannas	129,219	127,889	−1330	−1.03
9	Savannas	36,737	27,737	−9000	−24.50
10	Grasslands	9108	13,235	4127	45.31
11	Permanent Wetlands	3039	3880	841	27.67
12	Croplands	25,093	29,927	4834	19.26
13	Urban and Built-up Lands	38,558	38,966	408	1.06
14	Cropland/Natural Vegetation Mosaics	30,056	35,731	5675	18.88
16	Barren	135	115	−20	−14.81
17	Water Bodies	42,135	42,788	653	1.55

landscapes, such as farmland and water areas, have gradually decreased. In addition, urban buildings have been replaced, greatly changing the thermal radiation nature of the urban cushion surface. Therefore, studying the relationship between the urban thermal environment and the type of lower cushion surface has positive practical significance for rational planning of urban construction and the adoption of targeted mitigation of the heat island effect.

3.3.1. *Land use type*

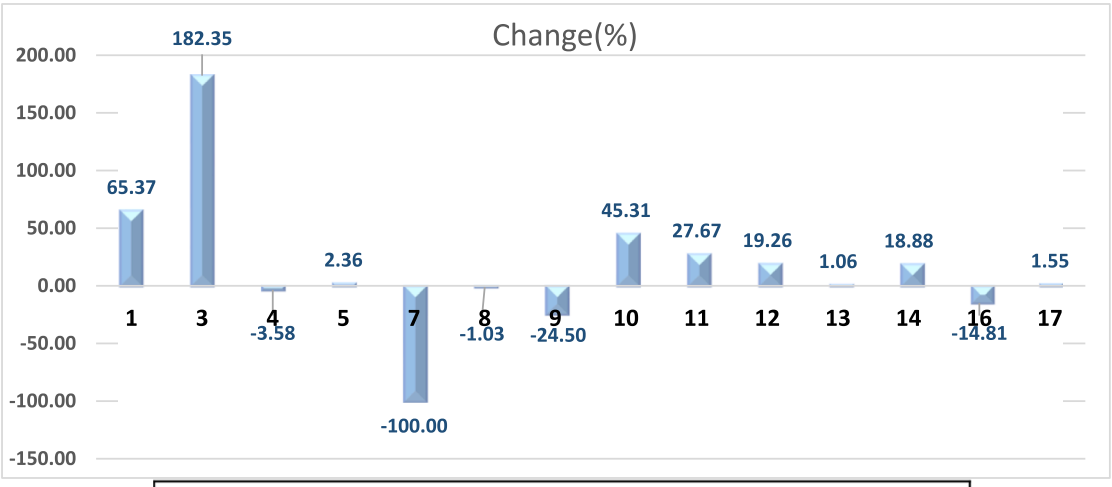
Since the surface temperature of the same thermal radiation in different land cover types is very different, the land use type greatly influences the change of urban surface temperature. There are many land use classification methods, most of which use three methods or combinations supervised classification, unsupervised classification, and artificial visual interpretation. This study uses land use data from New York in 2001 and 2020 on the MCD12Q1 satellite, which divides land use types into 18 types (Table 1). It analyzes the influence of land use types on urban surface temperature in combination with the surface temperature grade map of the corresponding temporal phases. Fig. 7 shows that in 2001 New York had 14 land types, while 2020, there were only 13 types. The type of Open Shrublands is missing.

Combining the land use data with the results of the same period of urban surface temperature data, it is found that there is a regularity between the land use type and the distribution of urban surface temperature levels. The surface temperature difference between different land use types is large. The distribution of urban areas coincides with the distribution of heat island areas and strong heat island areas, which is also due to the different geographical properties of the lower cushion surface of different land types. In order to obtain the area, change of land use type in more detail, data on land use type change are shown in Table 2 and visualized in Figs. 8 and 9.

As can be seen from the figure, from 2020 compared with 2001, mainly forest land was converted into farmland, and the urban area increased slightly. The growth rate was 1.06%. The decrease in the woodland area and the increase in farmland area have exacerbated the urban heat island effect to a certain extent.

3.3.2. *Topographical factors*

Due to its topographical reasons, it will affect the temperature distribution of urban heat fields to a certain extent. Therefore, this study utilizes digital elevation model (DEM) data from the study area, as seen



Legend:			
1	Evergreen Needleleaf Forests	10	Grasslands
3	Deciduous Needleleaf Forests	11	Permanent Wetlands
4	Deciduous Broadleaf Forests	12	Croplands
5	Mixed Forests	13	Urban and Built-up Lands
7	Open Shrublands	14	Cropland/Natural Vegetation Mosaics
8	Woody Savannas	16	Barren
9	Savannas	17	Water Bodies

Fig. 8. Proportional histogram of area change in land use type for 2001 and 2020.

in Fig. 10.

Combining the terrain with the surface temperature and comfort index can be analyzed that the surface temperature and elevation are inversely correlated, and, with the increase of elevation, the surface temperature shows a downward trend. However, in the area where the surface high-temperature heat field is more concentrated, the corresponding elevation value is low. It is also a residential and commercial dense area in the main urban area. People in this area from June to September are in a hot or even extremely hot feeling, which is extremely uncomfortable.

3.3.3. *Humanistic factors*

There are 62 cities in the state of New York. The largest city in the state and the most populous city in the United States is New York City. The New York agriculture industry is a major producer overall, ranking among the top five states for agricultural products, including maple syrup, apples, cherries, cabbage, dairy products, onions, and potatoes. In addition, the state is the largest producer of cabbage in the U.S. Huge capacity consumption, large population, and large tracts of land for agricultural production are the human factors leading to the obvious urban heat island effect in New York State.

4. Discussion

The existing human comfort research is more at the microscopic scale, the influence of the built environment on the human comfort index, and there are fewer macro-scale studies, and most of them are the exploration of methods(Coccolo et al., 2016; d'Ambrosio Alfano et al., 2014; Ji et al., 2022; Karimimoshaver and Shahrak, 2022). In a publication in 2022, the metropolitan surface urban heat island and urban thermal comfort were explored, and the influencing factors studied were land use/land cover change(Shahfahad et al., 2022). In this paper, we discussed the spatial and temporal changes and environmental impacts of human comfort by analyzing the changes in land surface temperature and land use on the spatiotemporal scale of the study area and considering the influence of geomorphology and other factors on thermal comfort.

Currently, the important causes of increasing heat field temperatures are, firstly, the type and nature of urban substrates, which along with urban expansion, the heat island effect will only intensify; the second is the mentioned anthropogenic heat, thus aggravating the heat island effect of the city. The third is the need to consume large amounts of non-renewable energy to supply the normal functioning of urban life. Therefore, in order to mitigate the urban heat island effect, firstly, based on the rational optimization of urban planning and construction and land use, we also apply the reduction of net radiation and anthropogenic heat emissions to effectively control the source of heat production and create a livable urban thermal environment. The following measures can be taken.

- Enhancement of greenery and watersheds

Strengthen the construction of greenery and water bodies, which play an important role as regulators in the urban ecosystem and are an important part of the ecosystem. They are also important parts of the ecosystem. These land types can not only effectively alleviate the urban heat island effect and reduce the impact of the heat island effect on the human body's thermal sensation but also can be used to improve the comfort of the human body. They can also beautify the city, and the urban green space system has now become one of the symbols of modern urban progress.

- Rational planning of urban layout and population distribution

New York City has become the most populous city in the United States. The development and urban expansion in New York State should be combined with New York State's topography, wind direction and other natural geography and atmospheric conditions, economic factors, and existing layout, etc., to reasonably plan the density and height of urban buildings.

- Improve the building's ability to regulate temperature

A study has shown that cool and green roofs can effectively alleviate

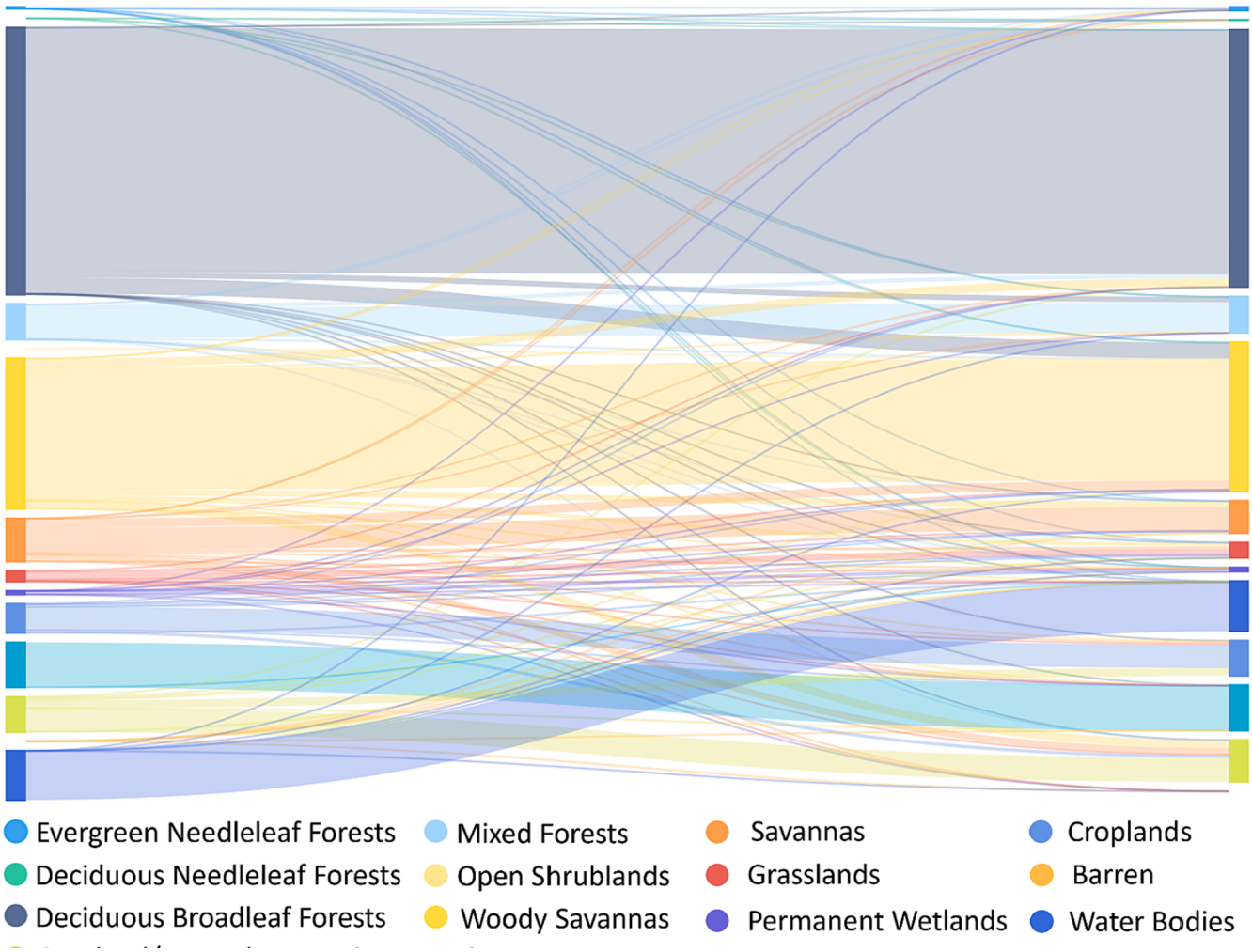


Fig. 9. Land Use Type Transfer Sankey Chart from 2001 to 2020.

the urban heat island effect and improve the thermal comfort of the human body(Wang et al., 2022). Reasonable design and modification of buildings can reduce heat reflection on hardened surfaces and improve ambient temperature, further improving environmental comfort.

The innovation of this paper lies in the use of multi-source data fusion, combined with the subjective comfort feelings of the human body, to deeply explore and study the urban heat field situation on the scale of time and space. However, due to the limitations of time, conditions, and other reasons, there are still some problems to be solved in this study, and the main problems and shortcomings are summarized as follows:

- o Only the analysis of people's heat discomfort, not the impact of cold discomfort on the human body.
- o The analysis of the influencing factors of the agglomeration effect of the urban thermal field is not comprehensive enough. The relationship between the spatial distribution of the urban thermal field and the land use cover type, topography, and meteorological factors of urban cushion surfaces needs to be discussed with the classification map with high classification accuracy.

There are many other factors that affect human comfort in their daily life. From a macro perspective, air quality will have an impact on human comfort; From the perspective of individual perceptions and micro-environmental experiences, such as ambient temperature, quality and quantity of light, noise level, spatial layout, time of a day, and personal cultural preferences, etc. Considering these spatio-temporal factors and sensory factors in environmental design and management can create comfortable spaces that promote well-being, productivity, and quality of life. However, due to the limitations of the size of one study, it is not possible to take into account many factors. This study mainly discusses the perception of human comfort from a macro temperature perspective. and we will add more influencing factors to future studies in order to obtain more comprehensive analysis results.

5. Conclusions

This paper takes New York and New Jersey as the study area. It uses GIS and RS technologies to analyze the human physical comfort level combined with urban heat island level using climate data from 2001 to 2020, with meteorological data, DEM data, and land use data for calculation. The variation patterns and spatial distribution characteristics of urban heat islands in New York and New Jersey urban agglomerations on time scales such as seasonal and interannual. The following are elaborated. The urban heat island phenomenon is significant in the New York urban agglomeration. On the temporal scale, the surface temperature in the 20 years from 2001 to 2020 showed an overall increasing trend, especially in July. On the horizontal spatial scale, regardless of the 12-month surface temperature variation, the New York City cluster is a concentration of relatively high temperatures, with a patchy pattern at low temperatures and a faceted pattern at high temperatures, especially in May and October, when the heat island effect is particularly pronounced. Most of the heat island areas in the study area are distributed in the New York City Cluster area, and combined with the

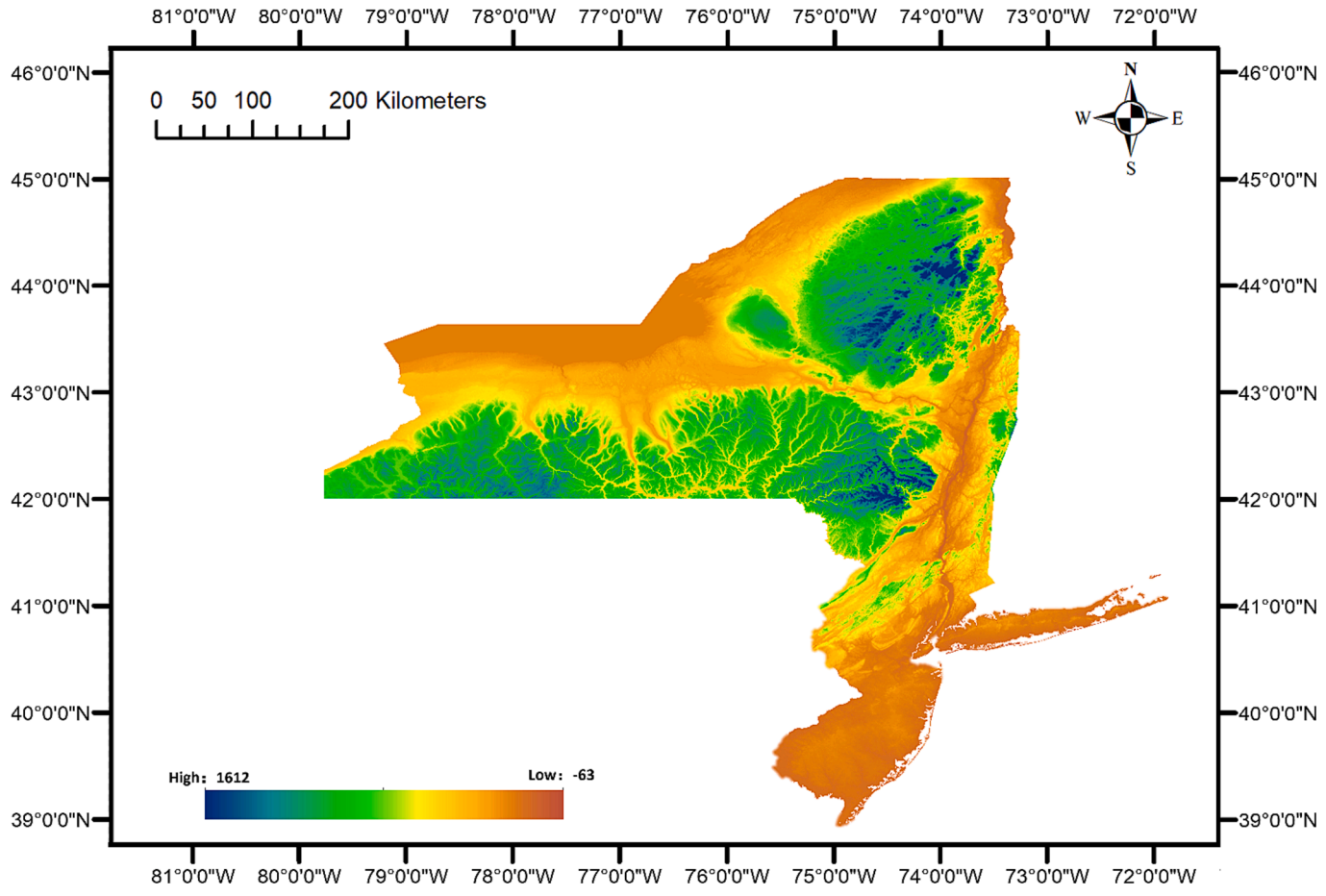


Fig. 10. DEM image of the study area.

thermal comfort map, it can be found that as the city expands with its development, people's DI index also increases, and the large area increases from a few feeling hot and uncomfortable to more than 50% feeling hot and uncomfortable. Due to the influence of the different nature of the substrates, the surface temperature difference between different parts of the city is also obvious. And except for the New York City cluster, other areas are mostly in a state of no thermal discomfort. The urban heat island is positively correlated with the low-level water substrates such as cement and asphalt and negatively correlated with the natural substrates such as water, soil, and vegetation; some low-temperature "valleys" of different sizes are formed on the surface of some green areas, water bodies and other areas, which can be seen that the existence of green areas and water bodies has a negative effect on the urban thermal environment. It indicates that over-concentrated urban development seriously affects residents' quality of life, and the heat island phenomenon should be appropriately shifted and reasonably greened to reduce the heat island phenomenon.

The results of this study can help city managers understand the human comfort index of the environment. The role played by other factors on both the scales of time and space were comprehensively considered and provided a reference for decision-making. Future research can combine deep learning methods with macro and micro scales to study the impact of the environment on human comfort and its coping strategies.

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The climate grid data used came from the website: https://www.worldclim.org/.

Land cover type data used were obtained from websites: https://modis.gsfc.nasa.gov/data/dataprod/mod12.php.

The land cover type data from the website: Application Status - EarthExplorer (usgs.gov).

Declaration of Competing Interest

The authors declare that they have no known competing financial interests or personal relationships that could have appeared to influence the work reported in this paper.

Data availability

Data will be made available on request.

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